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| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 09/29/2017 |
| **PI Name:** | Nicholson, Wayne Ph.D. | | |
| **Project Title:** | Comparative Evaluation of Microbial Transcriptomic Responses to Spaceflight Stress: Elucidating Underlying Molecular Mechanisms | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
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| **Zip Code:** | 32953 | **Congressional District:** | 3 |
| **Comments:** | NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13) | | |
| **Project Type:** | Ground,NASA GeneLab | **Solicitation / Funding Source:** | 2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System |
| **Start Date:** | 12/19/2016 | **End Date:** | 12/18/2019 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 12/18/2019 per NSSC information (Ed., 1/11/2019) | | |
| **Key Personnel Changes/Previous PI:** | October 2017 report: None | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX17AD51G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Microbial cells exposed to the human spaceflight environment are observed to respond differently to similarly-treated ground controls, a phenomenon dubbed the “spaceflight syndrome.” Over the past 4 decades, a number of experiments have been conducted inside various vehicles (e.g., Shuttle, Mir, ISS (International Space Station)) to understand how microbes sense and respond to stresses of human spaceflight including, but not limited to, microgravity and ionizing radiation. In early studies, various phenotypic outputs were measured, such as: growth rate and yield; virulence; biofilm formation and architecture; and resistance to antibiotics or abiotic stresses. With the advent of the genomics and post-genomics revolutions in biology, there have been initial efforts to understand more fundamental molecular aspects of the spaceflight syndrome, by performing global-scale "-omics" analyses of the transcriptome, proteome, metabolome, etc. of microbes exposed to spaceflight. Such studies have yielded valuable insights into the molecular responses of certain individual microbes to the spaceflight environment; furthermore, datasets from these experiments have been deposited into a common, publicly available database, the GeneLab Data System (GDS). We propose to use the data in the GDS as a starting point to explore commonalities and differences in the responses to spaceflight of various different microbes, with an eye towards uncovering possible common fundamental mechanisms.<br>Working Hypothesis: Exposure to the human spaceflight environment alters the physicochemical environment immediately surrounding bacterial cells and within the cells themselves (mainly due to microgravity effects and ionizing radiation). Cells sense the altered conditions and respond using the systems they possess--which evolved on Earth, not in space. A large proportion of microbial physiological responses are regulated at the level of transcription, thus the transcriptome is a logical place to search for common fundamental mechanisms.<br><br>Objectives: In order to address this hypothesis, we propose the following Specific Aims:<br><br>1. Mine the GDS bacterial transcriptome database and reprocess the data in a fashion that will allow cross-organism comparisons.<br><br>2. Search the spaceflight literature to identify additional relevant datasets that may have escaped inclusion in the GDS. Deposit these additional datasets in the GDS. Access and reformat the additional data as in Specific Aim 1.<br><br>3. Perform bioinformatics analyses to identify pathways and regulons significantly activated or repressed by exposure to spaceflight vs. ground controls in individual organisms.<br><br>4. Compare transcriptomes of the same organism grown in simulated microgravity (clinostats) vs. actual spaceflight conditions.<br><br>5. Perform comparisons across species, genera, families, and higher taxonomic groupings to find similarities and differences in spaceflight responses.<br><br>6. Develop specific hypotheses from the comparisons in Specific Aim 5 and perform wet-lab experiments to test these hypotheses in ground-based systems and in future spaceflight missions.<br><br>Expected significance: Microbes are ubiquitous in space vehicles, as well as on and within humans themselves. Because microbes play such important and essential roles in the health and well-being of astronauts and their life-support systems, it is imperative to understand their responses to the human spaceflight environment. The analyses proposed here will attempt to systematically identify common molecular mechanisms used by a wide range of bacteria to cope with the stresses encountered in human spaceflight, a primary goal of both the GeneLab endeavor and the broader mission of the NASA Space Biology Program. In addition, the pipelines and protocols developed in this study will have applications beyond GeneLab, and in principle can be applied to comparison of the transcriptomic responses of any collection of microbes to any given type of stress. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Microbes are ubiquitous in space vehicles, as well as on and within humans themselves. Because microorganisms play such important and essential roles in the health and well-being of astronauts and their life-support systems, it is imperative to understand their responses to the human spaceflight environment. The analyses performed in this study will systematically identify common molecular mechanisms used by a wide range of bacteria to cope with the stresses encountered in human spaceflight--a primary goal of both the GeneLab endeavor and the broader mission of the NASA Space Biology Program. In addition, the pipelines and protocols developed in this study will have applications beyond GeneLab, and in principle can be applied to comparison of the transcriptomic responses of any collection of microbes to any given type of stress. |
| **Task Progress:** | In the first year of this project, we have essentially accomplished Specific Aims 1-3.<br>Specific Aim 1. Mine the GeneLab bacterial transcriptome database and reprocess the data in a fashion that will allow cross-organism comparisons.<br><br>Graduate student Michael Morrison searched NASA’s GeneLab Data Repository and found complete transcriptome data for the following prokaryotic microorganisms grown in space vs. ground controls. These consisted of the organisms Rhodospirillum rubrum S1H (GLDS-31 and GLDS-39), Salmonella typhimurium (GLDS-11), Pseudomonas aeruginosa PAO1 (GLDS-15), and Bacillus subtilis (GLDS-138). Dataset GLDS-95, studying the effects of microgravity and antibiotics on Escherichia coli, was excluded from our analysis because it does not contain a control sample grown in the absence of antibiotic. These historical data sets had been generated using either microarrays or RNA-seq, and had been deposited in the GDS in a variety of different formats. Michael extracted the raw data sets and reprocessed them into a common format so that they could be further analyzed together.<br><br>Specific Aim 2. Search the spaceflight literature to identify additional relevant datasets that may have escaped inclusion in the GDS. Deposit these additional datasets in the GDS. Access and reformat the additional data as in Specific Aim 1.<br><br>From the literature, Michael identified a transcriptome data set from spaceflight vs. ground controls: Streptomyces coelicolor strain A3(2), deposited in the NIH Gene Expression Omnibus under accession number GSE53749. In a separate but related project, Michael has also generated transcriptome (RNA-seq) data from the Bacillus subtilis BRIC-21 flight (manuscript in preparation). As stated in the Specific Aim, both of these data sets will be deposited in the GDS.<br><br>Specific Aim 3. Perform bioinformatics analyses to identify pathways and regulons significantly activated or repressed |

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| | by exposure to spaceflight vs. ground controls in individual organisms.<br><br>As outlined in Specific Aim 1, Michael converted the disparate raw data sets into a common format meaningful for performing comparisons. Michael then developed a bioinformatic pipeline for data grooming and analysis of the data from all 7 data sets using the R package "limma." To date, differentially expressed genes have been identified and annotated with gene names, protein functions, and KEGG Orthology numbers using the UniProt database. Additionally, gene set enrichment analysis has been performed on all 7 datasets using the Broad Institute's multitool platform GenePattern. Significantly up- and down-regulated genes and pathways have been identified for each individual organism. |
| **Bibliography Type:** | Description: (Last Updated: 07/25/2025) |
| **Abstracts for Journals and Proceedings** | Morrison MD, Nicholson WL. "Mining the GeneLab Data System to identify common responses of bacteria exposed to spaceflight." Presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017.<br>33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. Abstract #78. , Oct-2017 |